

William Zhang

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EDUCATION

Purdue University (John Martinson Honors College)

West Lafayette, IN

Honors B.S. in Computer Science and Mathematics; GPA: 3.93/4.00

Expected May 2028

Coursework: Data Structures & Algorithms, Computer Architecture, Probability and Statistics, Programming in C, Linear Algebra, Real Analysis, Ordinary Differential Equations, Data Mining and Machine Learning, Web Information Search and Management, Systems Programming

EXPERIENCE

Machine Learning Intern

May 2026 – Aug 2026

Stesso AI

Portland, OR

- Built an OpenCV + PyTorch image-classification pipeline for home-inspection photos to predict repair categories, reaching 83% test accuracy on Kaggle benchmark data.
- Reduced token usage in a video-generation API workflow by 16%, lowering monthly inference costs while preserving output quality.

Project Lead and Full-Stack Developer

Sep 2025 – Present

Purdue Stack

West Lafayette, IN

- Led development of an Electron.js desktop app for Professor Xiaotao Yang's geology lab to process seismic data, cutting Jupyter-based analysis runtime by 34%.
- Contributed Python code to SeisGo, an open-source seismic analysis package on PyPI, adding Fourier-series functionality for research workflows.
- Built and shipped a mobile telehealth and appointment app with 8 engineers and designers for The Peer Network counseling clinic serving 700+ patients; released on the App Store and Google Play.

Research Intern

Jun 2024 – Jul 2025

University of Pennsylvania Perelman School of Medicine

Philadelphia, PA

- Worked in Professor Chamith Rajapakse's lab on ML research over 500,000+ emergency-room cases spanning 20 years of patient records.
- Developed a PyTorch/Keras neural-network pipeline to predict bone-fracture risk from environmental and movement features.
- Built an Arduino/C++ fitness-tracking device to quantify patient mobility and support physical-therapy outcome assessment.
- Co-authored an abstract and poster accepted to the 2025 New England Chapter of the American College of Sports Medicine conference.

PROJECTS

Finmath — Python, C/C++, pybind11, NumPy

Jan 2026 – Present

- Built Finmath with Boiler Quant, Purdue's quantitative finance club, as a Python package with a C/C++ backend for performance-critical workloads; published it on PyPI.
- Implemented time-series utilities and options-pricing models including Black-Scholes and binomial trees using pybind11 bindings.

Ecofriend — React Native, JavaScript, TypeScript, Java, Swift, Expo

May 2024 – Present

- Built a cross-platform mobile app for the 2024 Congressional App Challenge that helps natural-disaster victims find rebuilding resources.
- Released the app on the Apple App Store and Google Play Store.

TECHNICAL SKILLS & AWARDS

Languages: Java, Python, C/C++, JavaScript, TypeScript, R

Frameworks/Tools: React Native, Electron, Node.js, Next.js, Flask, Git, Docker, Linux, AWS, Jupyter

ML/Data: PyTorch, Keras, OpenCV, NumPy, pandas, SciPy, Matplotlib

Awards: USA Computing Olympiad (Gold Division, 2024 US Open Contest)